

Physiologic tooth form and The periodontium

It is assumed that the form of the teeth and their arrangement are related to incising or crushing food without causing damage to their supporting tissues (The periodontium).

Periodontium: Is the investing and supporting attachment system of teeth. **It consists of:**

Two soft tissues : Gingiva and Periodontal Ligaments

Two hard tissues : Cementum and Alveolar bone

The physiologic tooth form and arrangement may affect the periodontium either directly or indirectly.

The direct factors are: *****

- 1-Proximal contact areas.
- 2-Interproximal spaces.
- 3-Embrasures or spillways.
- 4-Facial and lingual contours of the crown.
- 5-Curvature of the cervical line.

2.The indirect factors are:

- 1-Cusp form.
- 2-Crown to Root ratio.
- 3- Root form.
- 4-Angulation of crown and root:
 - *Lingual angulations of the crown of lower posterior teeth.
 - *Mesial angulations of the crowns of permanent molars.
- 5- Self cleansing: Proper crown form allow self-cleansing ability of the tooth.

Direct factors

1- Proximal contact areas:

- *When *fully erupted*, the teeth have a tight *pointed contact* at their maximum width.
- *This pointed contact is transformed into area of contact by time **due to proximal attrition of the adjacent teeth . ****
- *Each tooth has *two* contact areas one *mesial* and one *distal* except the third molars.

The normal location of the contact areas are as follow:

On the individual tooth

The distal contact area has a more cervical location than the mesial one (except in the upper first premolar).

On all teeth

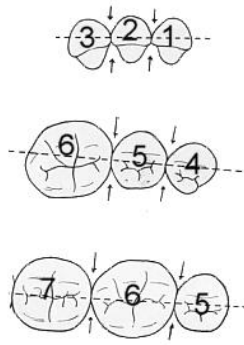
- *Contact areas become more cervically located in each quadrant.
- *The size of the contact areas increases in the same quadrant,



in *anterior teeth* Contact areas are

*centered in the faciolingual dimension.

*circumscribed



in *posterior teeth* contact areas are

*located slightly buccal to the center of the faciolingual dimension.

broad

Proximal contact position and size are greatly influenced by:

- 1- Length & width of crown.
- 2- Proximal wear.
- 4- Malocclusion.
- 5- Developmental anomalies.
- 6- Disproportion of the jaw and teeth size.
- 7- Extraction, unerupted or developmental missing teeth

Significances of proper proximal contact areas *****

- 1- Prevents interproximal food impaction and subsequently its stagnation with its harmful effect on the periodontium.
- 2- Protects the interdental gingiva from frictional trauma of food during mastication
- 3- Distributes the masticatory forces among the adjacent teeth in the individual dental arch.
- 4- Stabilize the dental arches by allowing each tooth to occlude with two opposing teeth (except lower central incisor and upper third molar).

Sequences of abnormal proximal contact area:

1- Food impaction between the teeth with consequent gingival inflammation which leads to destruction of the investing and supporting tissues and loss of the tooth.

2- Separation of teeth causes a change in the tooth alignment, that leads to shift in the masticatory forces to an angle that is not designed for the tooth to withstand.

This leads to occlusal trauma which, if prolonged, may cause destruction of the supporting tissues.

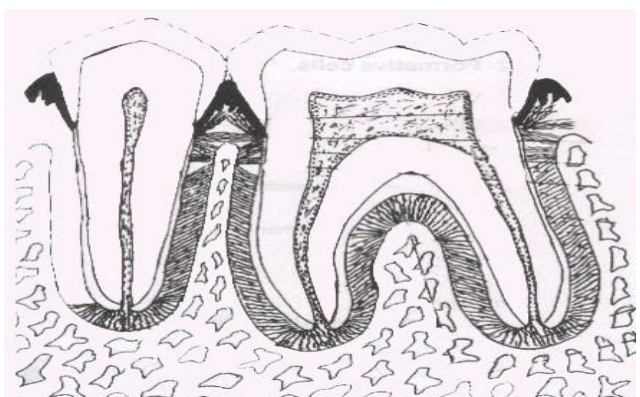
2- Interproximal spaces

These are triangular in shape found between the teeth .

The base of the triangle is the alveolar process, The sides are the proximal surfaces of contacting teeth and The apex of the triangle is the area of contact.

Significances of interproximal space

- ** provides space for the interdental gingival tissue. This tissue carries the blood vessels and nerves supply to all the investing tissues.
- ** provides considerable spacing for the alveolar bone between the neighboring teeth to support them in their sockets.



3- Embrasures or spillways

They are open spaces between the proximal surfaces of two adjacent teeth.

These spaces are triangular in shape.

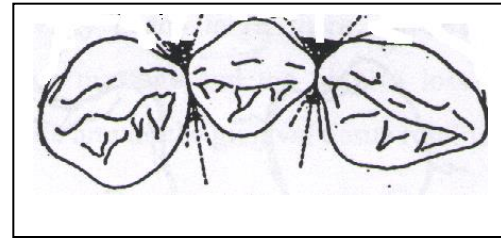
The size of the embrasure is determined by the location of the contact area...

The apex is located at the contact area and then diverges occlusally (incisally), buccally (labially) and lingually.

According to the location of the embrasure, there are three different types of embrasures:

- ** Labial embrasure
- ** buccal embrasure.
- ** Lingual embrasure.
- ** incisal & Occlusal embrasure.

The cervical embrasure is not exist, because it is filled by the interdental papillae . *****



Significance of Embrasures:

**It makes a *spillway* for the escapement of food during mastication thus reduces the masticatory forces applied on the teeth and also protects the gingiva.

**It makes the teeth able to do *self cleansing* action because the rounded, smooth surfaces of the crowns are more exposed to the cleansing action of foods, fluids and friction of tongue, lips and cheeks.

4- Facial and lingual contour

-In the anterior teeth:

The height of contour on both labial and lingual surfaces is located in the cervical third.

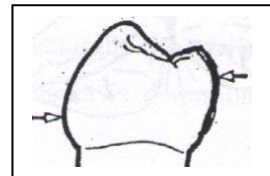
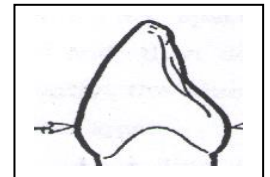
-In the posterior teeth:

the height of contour of the buccal surfaces is located in the cervical third.

On the lingual surface the height of contour is located at the middle third.

-It is well developed in the upper and ill developed in the lower teeth.

In the lower second premolar the height of contour lingually is located in the occlusal third.



Significance of proper facial and lingual contour :

**Hold the gingiva under definite tension.

**Deflect the food away from the gingival margin to prevent food impaction and gingival recession.

**Deflect the food over the gingiva which allow some massage to it .

Too small or absent convexity:

1-The food will push the gingival tissue apically leading to gingival recession.

2- As a result of this recession alveolar bone will be resorped.

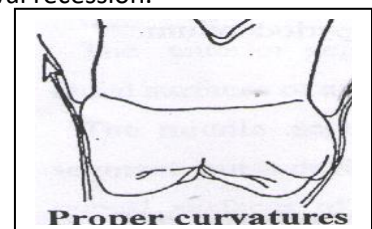
Too large convexity:

This will provide too much protection to the gingiva.

Gingiva will lose its tone.

Food will accumulate around the gingival region

resulting in its inflammation.

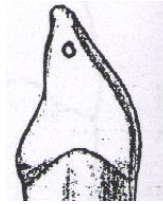


5-Curvature of the cervical line mesially and distally (Dentogingival junction)

Significance of cervical line curvature:

The importance of the cervical line curvature is due to the fact that it is the site of the attachment of the gingiva to the tooth by means of a structure called the attachment epithelium of the dentogingival junction.

*****In individual teeth** the curvature of the cervical line is greater mesially than distally.



***** In different teeth**, this curvature is greater in anterior than in posterior teeth.

Mesial

Distal

****In molars**

In the distal surfaces the curvature of the cervical line tends to be straight

